by far the leading explanation, on both theoretical and observational grounds.)

You might think that would be the end of it. Empty space has energy, we've measured it, cocoa and cupcakes all around.

But there's another question we're allowed to ask: What should we *expect* the vacuum energy to be? That's a funny question; since it's just a constant of nature, maybe we don't have the right to expect that it's any particular value at all. What we can do, however, is a quick-and-dirty estimate of how big we might guess the vacuum energy should be. The result is sobering.

The traditional way to estimate the vacuum energy is to distinguish between what the classical cosmological constant would be, and how quantum effects change that value. That's not really right; nature doesn't care that human beings like to start classically and build quantum mechanics on top of that. Nature is quantum from the start. But since all we're trying to do is get a very rough estimate, maybe this procedure is okay.

As it turns out, it's not okay. The quantum contribution to the vacuum energy is infinitely big. This kind of problem is endemic to quantum field theory; many calculations that we try to do by gradually including quantum effects end up giving us nonsensical, infinitely big answers.

But we shouldn't take those infinities too seriously. They can ultimately be traced to the fact that a quantum field can be thought of as a combination of vibrating modes at all different wavelengths, from incredibly long all the way down to zero. If we assume (for no especially good reason) that the classical minimum energy of each mode is zero, then the real-world vacuum energy is just the sum of all the additional quantum energies for each mode. Adding up the quantum energies for all those modes is what gives us an infinite vacuum energy. That's probably not physically realistic. After all, at very short distances we should expect spacetime itself to break down as a useful concept, as quantum gravity becomes impossible to ignore. It might make more sense to only include contributions with wavelengths larger than the Planck length, for example. We call this *imposing a cutoff*—looking at quantum field theory, but only including modes with wavelengths longer than a certain distance.

Unfortunately this doesn't quite fix the problem. If we estimate the quantum contribution to the vacuum energy by imposing a Planckscale